Your design is already very clean, practical, and well thought-out for powering vintage PCs (IBM PC/XT/AT class, Commodore, Amiga, etc.). It's compact, uses modern high-efficiency DC-DC modules, and includes nice touches like reverse-polarity protection and a power LED. Below are some **specific, actionable improvements and suggestions** ranging from small polish to more significant upgrades, depending on how far you want to take it.

1. Input Protection & Filtering (Highly Recommended)

- Add a **TVS diode** across the +12V input (after the barrel jack, before Q1).
  Suggested part: **SMAJ15A** or **P6SMB15A** (15 V clamp, handles automotive-load-dump-like transients from cheap laptop bricks).
- Add a **100 nF – 1 µF X7R ceramic** right at the input pins of every DC-DC module (you already have C1 for U2, but U1 and U3 are missing local ceramics). Switching regulators hate missing high-frequency bypass.
- Consider a small **common-mode choke** (e.g., 100 µH – 1 mH, 3–5 A) in series with +12V_IN if you ever see noise on the +5 V rail affecting logic.

2. Output Capacitance & Ripple

- The DFR0831 (+5 V module) is a buck converter and really wants **low-ESR output caps**.
  → Replace C2 (10 µF electrolytic) with **100–470 µF low-ESR electrolytic or polymer** + **10–22 µF MLCC** in parallel.
  Many 8-bit machines draw sharp 2–3 A pulses on +5 V; the stock 10 µF is marginal.
- The RECOM R-78CK5.0-0.5 (-5 V) is very clean, but still add **47–100 µF low-ESR** on -5 V instead of just 10 µF.
- Add **0.1 µF MLCC** directly at the output connector pins for +5 V and -5 V (decoupling right where the motherboard plugs in).

3. Reverse-Polarity MOSFET Q1 – Minor Improvements

- You're using a P-channel MOSFET correctly, but you're missing a **gate pulldown resistor** (100 kΩ from gate to source). Without it, if the barrel jack is unplugged the gate can float and Q1 may turn on partially.
- Add a **zener clamp** 15–18 V gate-source (or a 15 V zener + 1N4148 in series) to protect the gate oxide in case someone plugs in 24 V by mistake.

4. -12 V Rail (Murata NKE1212SC isolated module)

- You have L1 (22R686C wirewound inductor) in series with +12 V going to U2. That's unnecessary and actually hurts efficiency / regulation.
  → **Remove L1 completely** or replace with a 0 Ω link/ferrite bead.
- The NKE1212SC is only 83 mA. Most vintage PCs draw <30 mA on -12 V, so you're fine, but many motherboards expect -12 V within ±5 % from the moment +5 V comes up. The NKE module starts very slowly if C1 is too big.
  → 470 nF is actually perfect; don't increase it.

5. Connector & Pinout

Right now J2 looks like a custom 6-pin header. That's fine for a one-off, but for wider use consider one of these:

- Standard ATX 4-pin peripheral connector (Molex Mini-Fit Jr.) – widely available, keyed.
- Dual 6-pin IBM PC/XT-style (P8/P9) – exact drop-in replacement for original PSU.
- Or at least label the pinout clearly on the silkscreen (you already mostly do).

6. Power Good / Delay (Optional but Very Nice for Vintage Boards)

Many 286/386 and older boards expect the original IBM "Power Good" signal (goes high ~100–500 ms after +5 V is stable). Without it some boards stay in perpetual reset.
Easy fix: add a small **+5 V → Power-Good delay circuit** (TL431 + RC or a tiny microcontroller) and bring out a PG wire.

7. Minor Polish

- Move the LED current-limiting resistor note to schematic (or just use a plain 5 mm green LED + 3.3 kΩ–4.7 kΩ on +12 V).
- Add test points or 2.54 mm pads for +5 V, -5 V, -12 V, GND – makes probing easy.
- Add a silk-screen box with current ratings next to J2.

Summary – Quick Win List (do these first)

1. Add TVS diode on input (SMAJ15A)
2. Add 100–470 µF low-ESR + 10 µF ceramic on +5 V output
3. Add 100 kΩ pulldown on Q1 gate
4. Remove L1 (or replace with ferrite bead)
5. Add 0.1 µF ceramics right at J2 pins for +5 V and -5 V

With those changes this becomes a bulletproof, low-noise, drop-in modern replacement for basically any 1980s–early-1990s PC or peripheral that originally used a linear PSU.

Great work overall — this is already one of the cleaner “modern PSU for vintage iron” designs I’ve seen!

↳ Power Good signal details

↳ ATX power supply differences

↳ Include updated schematic